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Course: Calculus Beta

Assignment: MyLab Math - Uge 3

Evaluate the integral using integration by parts.

$$\int x^2 \cos(2x) \, dx$$

To evaluate $\int x^2 \cos(2x) \, dx$, use integration by parts. Use the functions $u = x^2$ and $dv = \cos(2x) \, dx$ to integrate using the form $\int u \, dv = uv - \int v \, du$.

For $u = x^2$, find the expression for du .

$$du = 2x \, dx$$

For $dv = \cos(2x) \, dx$, find the expression for v .

$$v = \frac{1}{2} \sin(2x)$$

Substitute these expressions into the formula for integration by parts.

$$\begin{aligned} \int u \, dv &= uv - \int v \, du \\ \int x^2 \cos(2x) \, dx &= (x^2) \left(\frac{1}{2} \sin(2x) \right) - \int \left(\frac{1}{2} \sin(2x) \right) 2x \, dx \\ &= \frac{1}{2} x^2 \sin(2x) - \int x \sin(2x) \, dx \end{aligned}$$

Note that the second term in the result is another integral that can be integrated by parts. To integrate $\int x \sin(2x) \, dx$ by parts, use $u = x$ and $dv = \sin(2x) \, dx$. Find du and v .

$$du = dx$$

$$v = -\frac{1}{2} \cos(2x)$$

Substitute these expressions into the formula for integration by parts.

$$\begin{aligned} \int u \, dv &= uv - \int v \, du \\ \int x \sin(2x) \, dx &= (x) \left(-\frac{1}{2} \cos(2x) \right) - \int \left(-\frac{1}{2} \cos(2x) \right) dx \\ &= -\frac{1}{2} x \cos(2x) + \frac{1}{2} \int \cos(2x) \, dx \end{aligned}$$

Evaluate the remaining integral. Omit the arbitrary constant of integration until the original integral is evaluated.

$$\begin{aligned} \int x \sin(2x) \, dx &= -\frac{1}{2} x \cos(2x) + \frac{1}{2} \int \cos(2x) \, dx \\ &= -\frac{1}{2} x \cos(2x) + \frac{1}{4} \sin(2x) \end{aligned}$$

Substitute this expression for $\int x \sin(2x) \, dx$ into the expression for the original integral, $\int x^2 \cos(2x) \, dx$. Note the arbitrary constant of integration, used here to account for all constant terms resulting from previous integrations.

$$\begin{aligned} \int x^2 \cos(2x) \, dx &= \frac{1}{2} x^2 \sin(2x) - \int x \sin(2x) \, dx \\ &= \frac{1}{2} x^2 \sin(2x) + \frac{1}{2} x \cos(2x) - \frac{1}{4} \sin(2x) + C \end{aligned}$$